Control Engineering Practice 108 (2021) 104721

Contents lists available at ScienceDirect

Control Engineering Practice

journal homepage: www.elsevier.com/locate/conengprac

MUDE-based control of quadrotor for accurate attitude tracking

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ARTICLE INFO

Keywords:
Quadrotor control
High-accuracy attitude tracking
Actuator model identification
Disturbance rejection

ABSTRACT

In this paper, a modified uncertainty and disturbance estimator (MUDE)-based attitude controller for quadrotors based on a high-accuracy actuator dynamic model is proposed. The actuator dynamics are approximated by a first-order plus time-delay dynamic model, and both static and dynamic models of the actuator are identified using a motor test platform. The actuator model is further exploited as a virtual sensor to provide the actuator-generated torque estimates for the controller. By these estimates, an MUDE is constructed to compensate for not only the uncertainties and disturbances in attitude dynamics but also the time delay and disturbances in actuator dynamics. The systematic consideration of actuator disturbance rejection significantly improves the overall attitude tracking performance during maneuvers, which is clearly distinguished from other existing solutions assuming ideal actuators. Furthermore, the stability and performance analysis of the closed-loop system is presented, showing that the estimation error and the closed-loop tracking error can be reduced by tuning a single parameter. Finally, the proposed control, as well as classic uncertainty and disturbance estimator-based control and cascade PID controller, are implemented and compared on a real quadrotor platform. The experimental results show the effectiveness and advantage of the proposed control method with respect to signal tracking and disturbance rejection.

1. Introduction

In recent years, quadrotors have received notable attention from both practical engineering and academic communities for their potential prospects in various applications (Amovlab, 2020; Bo, Xian, Yao, & Xu, 2015; Falanga, Kleber, Mintchev, Floreano, & Scaramuzza, 2018; Lee & Kim, 2017; Mahony, Kumar, & Corke, 2012). A class of quadrotor systems called industrial-grade quadrotors have been designed to target a wide range of industrial application scenarios, such as package delivery, plant protection, and fire control. There are two significant challenges in the control of these drones compared with that of commercial-grade quadrotors. One is to achieve precise trajectory-tracking during possibly fast maneuvers to perform complicated missions, and the other is to handle the system uncertainties and external disturbances in complex working environments.

Attitude control is one of the critical and fundamental problems for industrial-grade quadrotors, but achieving high-accuracy tracking performance is by no means simple. In most existing works about quadrotor attitude control (such as Bo et al., 2015; Falanga et al., 2018; Lee & Kim, 2017; Mahony et al., 2012), the actuator is assumed to be an ideal system. However, the actuator subsystem contains a transient response in reality (Derafa, Madani, & Benallegue, 2007; Madani & Benallegue, 2007), rendering the actuator dynamics not negligible when considering (aggressive) maneuvers. Hence, the overall control performance may be significantly deteriorated if actuator dynamics are not systematically taken into account (Liu, Yuan, Yu, & Zhang, 2017). In the context of this paper, the consideration of quadrotor actuator dynamics in the attitude control design is practically meaningful, since the actuators, including the electronic speed controllers (ESC), motors and propellers, are the main contributor to the quadrotor dynamics. However, relevant discussions on high-performance control designs by exploiting high-accuracy actuator dynamics are few in the literature. Existing works in which the actuator dynamics are considered mostly model or approximate the actuator as a simple first-order system (Awan, 2014; Faessler, Falanga, & Scaramuzza, 2017; González I. Salazar, Torres, Lozano, & Romero, 2013; Liu et al., 2017; Lu, 2018; Mokhtari & Abdelaziz, 2016; Phang, Li, Wang, Chen, & Lee, 2014). In Liu et al. (2017), a fault-tolerant tracking control (FTTC) method with application to an unmanned quadrotor in the presence of actuator faults is proposed, where a first-order actuator dynamics with limited operating regions are used in the controller design. By feeding back an augmented state (the actuator-generated torque) of the system,

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https://doi.org/10.1016/j.conengprac.2020.104721
Received 28 April 2020; Received in revised form 10 November 2020; Accepted 24 December 2020
Available online 6 January 2021
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a linear quadratic regulator (LQR)-based controller is proposed in Faessler et al. (2017) to deliver accurate body-rate trajectory-tracking during aggressive flights. In González I. Salazar et al. (2013), a two-loop attitude stabilization controller is designed based on motor speed feedback, and certain robustness is observed from outdoor experiments in hovering mode. Works (Awan, 2014; Mokhtari & Abdelaziz, 2016) adopt a first-order linearized actuator model to design an output feedback altitude–attitude controller and a passivity-based simple adaptive control strategy. In Lu (2018), a detailed actuator dynamic model is built, and a set of identified results describing the relationship among PWM signal, motor speed, and battery are presented, but the actuator model is not exploited in the designed cascade PID-like attitude controllers. Work (Phang et al., 2014) presents a comprehensive model of brushed motor dynamics and model identification results, by which good tracking performance is delivered even only using a simplified static actuator model-based controller. Nevertheless, the topic of fully utilizing the actuator model to improve the quadrotor attitude tracking accuracy is still open, since the questions like how comprehensive and accurate the actuator model should be, or whether the use of the static model is sufficient, have not yet been clearly answered.

Meanwhile, the robustness with respect to various disturbances and model uncertainties is of practical importance for industrial-grade quadrotors. In the last decade, numerous robust control methodologies have been applied to quadrotors, such as PID (Bouabdallah, Noth, & Siegwart, 2004), adaptive control (Lee, 2012), uncertainty and disturbance estimator (UDE) (Lu, Ren, & Parameswaran, 2020; Lu, Ren, Parameswaran, & Zhong, 2018), active disturbance rejection control (ADRC) (Yang, Cheng, Xia, & Yuan, 2017; Zhao, Zhang, Ma, & Xia, 2019), backstepping control (Jia et al., 2017; Jiang, Lin, & Song, 2018), sliding-mode control (Hou, Lu, & Tu, 2020; Jiang, Song, & Lin, 2019), and disturbance-observer-based control (DOBC) (Wang & Su, 2015). In Lu et al. (2020), the fusion of the UDE technique with the unit quaternion is applied in the trajectory tracking control of quadrotors, and flight tests are conducted to demonstrate the global singularity-free tracking property and the superior robustness of the proposed controller. The perturbations of gust winds are considered and estimated by an extended state observer in Yang et al. (2017), and a dual closed-loop control framework using ADRC and proportional–derivative control is implemented therein to achieve disturbance rejection. In Alexis, Nikolakopoulos, and Tzes (2011), a switching model predictive attitude controller based on a piecewise affine (PWA) model of the quadrotor's attitude dynamics is presented to handle the atmospheric disturbances, and the experimental results indicate the high overall efficiency of the proposed control scheme under wind gust disturbances. A robust inner-loop controller for the quadrotor stabilization based on nonlinear H-infinity theory is designed in Raffo, Ortega, and Rubio (2010), which is able to reject sustained disturbances by the use of the integral action in the state vector. Despite the differences in the design philosophies, none of the aforementioned works has considered the adverse effects brought by actuators in their disturbance rejection mechanisms. Assuming ideal actuators makes the control problem simpler, but the disturbance rejection performance can be largely degraded by the neglect of the actuators' dynamic responses because the quadrotor motion and actuator dynamics are in the same time-scale (Lu, 2018). This point is supported by the results presented in work (Castillo et al., 2019), where the actuator dynamics are considered in the design of the disturbance observer to improve the disturbance estimation performance during aggressive maneuvers. However, the time delay and disturbances on the actuators are not resolved therein, which in practice may lead to control performance degradation or even system instability (Wu & Park, 2019).

To the best of our knowledge, the idea of utilizing a high-accuracy actuator dynamic model in the control design to simultaneously improve the attitude tracking and disturbance rejection performance is proposed in this paper for the first time. The main purpose of this work is to develop a simple and effective control solution for quadrotor attitude control, which is capable of not only delivering high-accuracy tracking performance during (aggressive) maneuvers but also compensating for vast model uncertainties and external disturbances with respect to both attitude motion and actuator dynamics. To this end, the quadrotor attitude and actuator dynamic models are built firstly, and the parameters of the actuator static and dynamic model are identified using a motor test platform. The time delay of the actuators is experimentally verified to be non-negligible, and thus a first-order plus time-delay (FOPDT) model, instead of a simple first-order model considered in Awan (2014), Castillo et al. (2019), Faessler et al. (2017), González I. Salazar et al. (2013), Liu et al. (2017) and Mokhtari and Abdelaziz (2016), is adopted to describe the actuator dynamics and is further utilized for the nominal control design and disturbance estimation. Meanwhile, the static model serves as a mathematical mapping between thrust and throttle commands, and it is used in the thrust mixing module in the quadrotor flight controller. A modified uncertainty and disturbance estimator (MUDE) is developed to estimate and compensate for not only the lumped disturbances on the attitude and actuator dynamics but also the actuator time delay. The proposed control is implemented and validated on a real quadrotor platform. Compared with the cascade PID controller used in most commercial autopilots (Dronecode, 2020) and classic UDE-based control presented in Lu et al. (2018), the proposed control delivers smaller overshoot for step input, better tracking accuracy for time-varying input, and faster performance recovery ability under disturbances, according to the experiment results. The main contributions of this paper are summarized as follows:

(1) A high-accuracy actuator dynamic model is exploited to enhance nominal feedback control performance, which is clearly distinguished from works (Bouabdallah et al., 2004; Hou et al., 2020; Jia et al., 2017; Jiang et al., 2018, 2019; Lee, 2012; Lu et al., 2020, 2018; Wang & Su, 2015; Yang et al., 2017; Zhao et al., 2019) assuming ideal actuators and works (Awan, 2014; Castillo et al., 2019; Faessler et al., 2017; González I. Salazar et al., 2013; Liu et al., 2017; Lu, 2018; Mokhtari & Abdelaziz, 2016; Phang et al., 2014) using simplified or static actuator models. The actuator model is further used as a virtual sensor to provide estimates of actuator-generated torques for the feedback controller, which helps to enlarge the system stability margin and improve the convergence speed of attitude tracking, thus improving the attitude tracking accuracy during maneuvers.

(2) Systematic consideration of the disturbed actuator dynamics is taken in the design of MUDE to improve the overall disturbance rejection capability. Specifically, the virtual sensor is additionally employed in MUDE to obtain a better disturbance estimation, whereas existing approaches (Bouabdallah et al., 2004; Hou et al., 2020; Jia et al., 2017; Jiang et al., 2018, 2019; Lee, 2012; Lu et al., 2020, 2018; Wang & Su, 2015; Yang et al., 2017; Zhao et al., 2019) avoid using immeasurable actuator states in the disturbance rejection mechanisms.

(3) A novel idea of separately designing the state feedback controller and the MUDE in two different coordinates is proposed to resolve the mismatched disturbance issue caused by the consideration of actuator dynamics. A nonsingular coordinate transformation is introduced to build an auxiliary system with equivalent matched disturbance, and the feedback controller is designed in the new coordinate.

The rest of this paper is organized as follows. Section 2 presents the quadrotor dynamic model and the actuator model. Section 3 presents the proposed MUDE attitude control law, and the stability analysis is given in . The experimental setup and obtained results are given in Section 5. Finally, Section 6 concludes this paper.

2. Quadrotor system modeling

In this section, the quadrotor attitude and actuator models used for the controller design are presented. First, the attitude dynamic model is built and linearized using feedback linearization techniques. Second, the actuator model is introduced, and the identification results are presented.

2.1. Attitude dynamic model

As shown in Fig. 1, the quadrotor platform with an X airframe configuration is used in this work, and it is controlled by four motors that are mounted on the end of the arm. Euler angles $\boldsymbol{\Theta} = [\phi, \theta, \psi]$ are used to express the orientation of the quadrotor, where ϕ is the roll angle, θ is the pitch angle, and ψ is the yaw angle, respectively. By applying the so-called small angle approximation (Lu et al., 2018), the attitude dynamics of the quadrotor in the inertial frame can be described as

$$\begin{aligned}
\ddot{\phi} &= \frac{1}{J_{xx}}\left[\left(J_{yy}-J_{zz}\right)\dot{\theta}\dot{\psi} + J_r\overline{\Omega}\dot{\theta} + d_{\tau x} + \tau_x\right],\\
\ddot{\theta} &= \frac{1}{J_{yy}}\left[\left(J_{zz}-J_{xx}\right)\dot{\phi}\dot{\psi} - J_r\overline{\Omega}\dot{\phi} + d_{\tau y} + \tau_y\right],\\
\ddot{\psi} &= \frac{1}{J_{zz}}\left[\left(J_{xx}-J_{yy}\right)\dot{\phi}\dot{\theta} + d_{\tau z} + \tau_z\right],
\end{aligned} \tag{1}$$

where $\boldsymbol{J} = \mathrm{diag}([J_{xx}, J_{yy}, J_{zz}])$ is the moment of inertia of the quadrotor, J_r denotes the inertia of propeller and rotor in z-axis. Define $\overline{\Omega} = \varpi_1 + \varpi_2 - \varpi_3 - \varpi_4$, where ϖ_i is the rotation speed of ith motor. $\boldsymbol{d}_\tau = \begin{bmatrix} d_{\tau x} & d_{\tau y} & d_{\tau z}\end{bmatrix}^T$ is the disturbance torque, and if the quadrotor is not strictly symmetric in practice, i.e. the inertial products J_{xy}, J_{yz} and J_{zx} are not zero, the coupled dynamics are also lumped in this term. $\boldsymbol{\tau} = \begin{bmatrix}\tau_x & \tau_y & \tau_z\end{bmatrix}^T$ is the torque generated by the four motors and can be computed as

$$\begin{aligned}
\tau_x &= \frac{\sqrt{2}}{2} l\left(-T_1 + T_2 + T_3 - T_4\right),\\
\tau_y &= \frac{\sqrt{2}}{2} l\left(T_1 - T_2 + T_3 - T_4\right),\\
\tau_z &= \frac{c_M}{c_T}\left(T_1 + T_2 - T_3 - T_4\right),
\end{aligned} \tag{2}$$

where l is the quadrotor's arm length, c_M and c_T are actuator parameters that will be identified in the next subsection, and T_i is the actuator-generated thrust from the ith actuator.

In order to decouple and simplify the dynamics (1), the feedback linearization techniques are utilized. Define virtual inputs as

$$u_x = \frac{\tau_x}{J_{xx}}, \quad u_y = \frac{\tau_y}{J_{yy}}, \quad u_z = \frac{\tau_z}{J_{zz}}, \tag{3}$$

and define the lumped disturbances as

$$\begin{aligned}
f_\phi &= \frac{1}{J_{xx}}[(J_{yy}-J_{zz})\dot{\theta}\dot{\psi} + d_{\tau x} + J_r\overline{\Omega}\dot{\theta}],\\
f_\theta &= \frac{1}{J_{yy}}[(J_{zz}-J_{xx})\dot{\phi}\dot{\psi} + d_{\tau y} - J_r\overline{\Omega}\dot{\phi}],\\
f_\psi &= \frac{1}{J_{zz}}[(J_{xx}-J_{yy})\dot{\theta}\dot{\phi} + d_{\tau z}].
\end{aligned} \tag{4}$$

Then the dynamics (1) can be rewritten into the compact form of

$$\ddot{\boldsymbol{\Theta}} = \boldsymbol{u} + \boldsymbol{f}, \tag{5}$$

where $\boldsymbol{u} = \begin{bmatrix}u_x & u_y & u_z\end{bmatrix}^T$, $\boldsymbol{f} = \begin{bmatrix}f_\phi & f_\theta & f_\psi\end{bmatrix}^T$.

Remark 1. In order to reduce the effects of model uncertainties on the control performance, the coupling terms $(J_{yy}-J_{zz})\dot{\theta}\dot{\psi}$, $(J_{zz}-J_{xx})\dot{\phi}\dot{\psi}$, and $(J_{xx}-J_{yy})\dot{\theta}\dot{\phi}$ in the quadrotor dynamics (1) are included in the lumped disturbances terms expressed in (4), and they will be estimated and compensated dynamically by the proposed MUDE-based control. If these coupling terms can be precisely known or approximated within a reasonable margin, they can be incorporated by the virtual inputs (3) and be canceled by their corresponding compensation terms via the feedback linearization technique, while the residual modeling error will be included in the lumped disturbances (4).

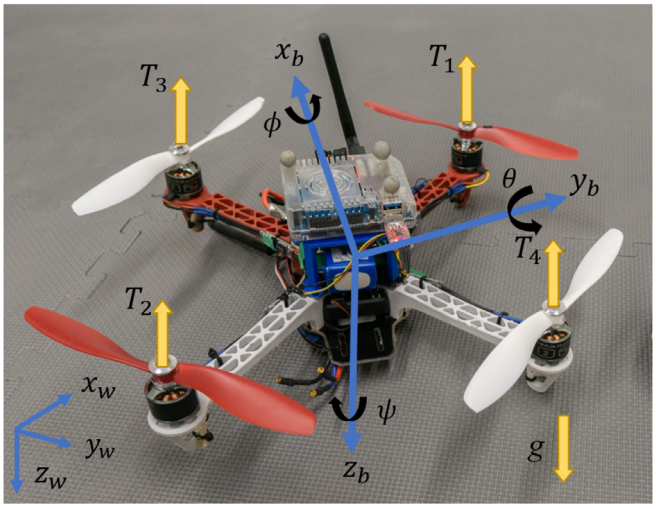


Fig. 1. Quadrotor with coordinate systems.

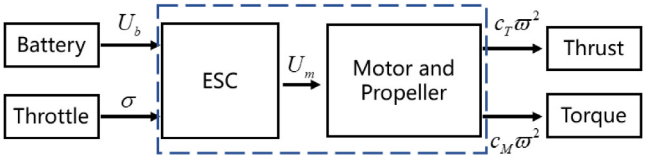


Fig. 2. The actuator model.



Fig. 3. Series 1580 Thrust Stand.

2.2. Actuator model and parameter identification

As shown in Fig. 2, each actuator of a quadrotor consists of an ESC, a motor, and a propeller. With the input throttle σ_i for each motor and the battery voltage U_b, the motor is driven by an ESC, which converts the throttle command into multiple out-of-phase voltage outputs. The spinning motor produces the force T_i and reactive torque τ_i with the appropriate propeller described by the following equations:

$$T_i = c_T\varpi_i^2, \quad \tau_i = c_M\varpi_i^2, \tag{6}$$

where c_T and c_M are the thrust constant and torque constant respectively.

To identify the parameters of the actuator model, Series 1580 Thrust Stand from RCbenchmark (RCbenchmark, 2020) is adopted as shown in

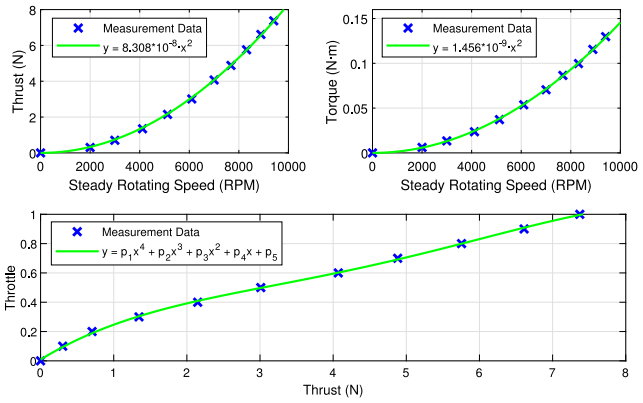


Fig. 4. Identification results of actuator static model.

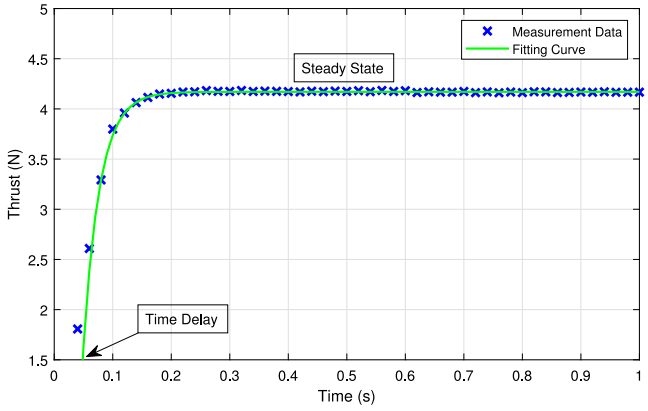


Fig. 5. Curve fitting result of actuator dynamic model.

Fig. 3. The throttle command, thrust, torque, voltage as well as motor rotation speed can be measured and recorded at the same time.

2.2.1. Static model identification

Under the same experimental conditions (the normal atmospheric temperature and battery voltage in the working range), different throttle commands in the range of $[0, 1]$ are given and the results when the motor arrives at the steady-state are recorded. Importing the experiment data into the Matlab curve fitting Toolbox, the static model identification results are obtained, as shown in Fig. 4.

From Fig. 4, it is observed that the relationship between the thrust/torque and rotation speed satisfies the quadratic equations (6). Then the motor parameters can be obtained from the fitted curves: $c_T = 8.308 \times 10^{-8}$ N/RPM^2, $c_M = 1.456 \times 10^{-9}$ N m/RPM^2.

To map the thrust commands T_i (control inputs obtained from controller) to throttle commands σ_i (low level commands in the autopilot), the following fourth-order polynomial approximation is adopted:

$$\sigma_i = p_1 {T_i}^4 + p_2 {T_i}^3 + p_3 {T_i}^2 + p_4 T_i + p_5, \tag{7}$$

where $p_1 = -0.00069$, $p_2 = 0.01271$, $p_3 = -0.07948$, $p_4 = 0.30521$, $p_5 = 0.00878$.

2.2.2. Dynamic model identification

The static model only gives a simple approximation of the motor dynamics in terms of a static command. However, in the dynamic responses of motors, the time delay and response time need to be systematically considered. Instead of building a comprehensive model (considering electrical, mechanical and aerodynamic properties) for the actuators as Phang et al. (2014), the model is simplified into the following first-order plus time-delay form

$$T = \frac{e^{-\tau_0 s}}{\alpha s + 1} T_{ss}, \tag{8}$$

where τ_0 is the time delay, α is the time constant and T_{ss} is the steady state of the force.

Under the same experimental conditions, a step command is applied and the experimental data are recorded for each test. From the curve fitting results shown in Fig. 5, it is seen that the time delay in the actuator response is significant, and the response process is close to an FOPDT model.

To improve the accuracy of the identification results, this experiment is repeated 8 times with different initial and final input throttles, which mostly cover the working range of the motors. The results are shown in Table 1.

From these results, it is seen that the time delay of the actuator always exists. Since the control frequency of attitude control is only 250 Hz (0.004 s per control period) in most autopilots, the actuator time delay cannot be neglected in practice. Averaging the test results gives

$$\alpha = 0.035, \quad \tau_0 = 0.04 \text{ s}. \tag{9}$$

Substitute (8) into (2), the dynamic model of the actuator in time domain is given as

$$\dot{\boldsymbol{\tau}}(t) = -\frac{1}{\alpha}\boldsymbol{\tau}(t) + \frac{1}{\alpha}\boldsymbol{u}_\tau\left(t - \tau_0\right) + \boldsymbol{d}_\tau, \tag{10}$$

where $\boldsymbol{u}_\tau = \begin{bmatrix} u_\phi & u_\theta & u_\psi \end{bmatrix}^T$ are the torque commands, and $\boldsymbol{d}_\tau$ denotes the lumped disturbances on the actuators, such as battery voltage drop and ground effects.

In real implementations, the nominal actuator model, i.e. Eq. (10) with $\boldsymbol{d}_\tau = \boldsymbol{0}$, is exploited as a virtual sensor to estimate the motor-generated torque. With good torque estimations, the inner-loop controller can be designed to achieve higher-accuracy control of the attitude by a systematic consideration of nonideal actuators.

Table 1
Identification results of actuator dynamic model.

Experiment No.	1	2	3	4	5	6	7	8
Initial throttle	0.3	0.2	0.2	0.2	0.3	0.4	0.4	0.5
Desired throttle	0.6	0.3	0.4	0.5	0.5	0.5	0.6	0.6
τ_0 (s)	0.0400	0.0545	0.0207	0.0729	0.0195	0.0396	0.0283	0.0479
α	0.0285	0.0292	0.0523	0.0249	0.0510	0.0319	0.0334	0.0301

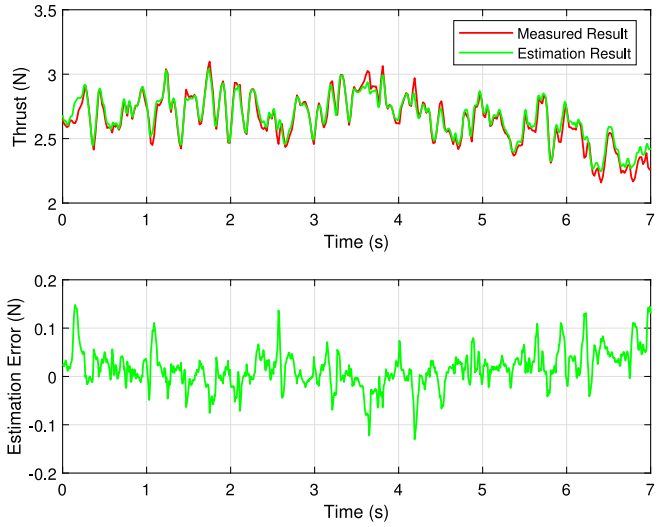


Fig. 6. Validation results for the identified actuator model.

Remark 2. The identification results shown in Table 1 and Eq. (9) are only valid for the specific actuator system (including the motor, ESC and battery) introduced in Section 5. If any component of the actuator is changed, the identification needs to be redone. The identification experiment is conducted under the same experimental conditions: normal atmospheric temperature (25.0 °C) and working range battery voltage (11.1 ∼ 12.6 V). Within reasonable ranges of experimental condition changes, the identification results are meaningful. To verify the identification results, the same torque command u_τ logged from a real flight is applied to both the Thrust Stand and the identified nominal actuator model, and then the thrust measurement from the testbed and thrust estimate using the model are compared. Note that the thrust estimation is calculated by $T = \frac{\tau}{L}$, where τ is the estimated torque and $L = 0.165$ m is the length of the quadrotor arm. The comparison results are shown in Fig. 6, from which it is seen that the estimation error is about ±0.1 N. The nominal actuator model is of high-accuracy in the sense that the validation results only exhibit an estimation error with a 4% amplitude of the actual actuator-generated thrust.

3. MUDE controller design

A linear time invariant (LTI) model for the full inner-loop dynamics (including the attitude and actuator dynamics) can be obtained by combining (1) and (10) as

$$\begin{bmatrix}\dot{\boldsymbol{\Theta}}\\ \ddot{\boldsymbol{\Theta}}\\ \dot{\boldsymbol{\tau}}\end{bmatrix}=\begin{bmatrix}\boldsymbol{0} & \boldsymbol{I}_3 & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{J}^{-1}\\ \boldsymbol{0} & \boldsymbol{0} & -\frac{1}{\alpha}\boldsymbol{I}_3\end{bmatrix}\begin{bmatrix}\boldsymbol{\Theta}\\ \dot{\boldsymbol{\Theta}}\\ \boldsymbol{\tau}\end{bmatrix}+\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{0}\\ \frac{1}{\alpha}\boldsymbol{I}_3\end{bmatrix}\boldsymbol{u}_\tau+\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{J}^{-1}\boldsymbol{f}_1\\ \boldsymbol{f}_2\end{bmatrix}, \tag{11}$$

where $\boldsymbol{I}_3$ is the 3×3 identity matrix, $\boldsymbol{f}_1 = \boldsymbol{f}$ is the lumped disturbance torque, and $\boldsymbol{f}_2 = \frac{1}{\alpha}\left[\boldsymbol{u}_\tau\left(t-\tau_0\right)-\boldsymbol{u}_\tau(t)\right]+\boldsymbol{d}_\tau$ is the lumped disturbance on the actuators. According to Lu et al. (2018), Zhao et al. (2019) and Zhu, Liu, and Li (2015), the following assumption on the lumped disturbances is employed.

Assumption 1. The disturbances $\boldsymbol{f}_1$, $\boldsymbol{f}_2$ and the derivative $\dot{\boldsymbol{f}}_1$ are bounded for all $t \geq 0$.

In the following, the controller for each channel will be designed separately. Note that the dynamics of the three channels are in the same form given by (11). Therefore, without loss of generality, a controller for the pitch channel is designed, and then the same controller structure with different control gains (in accordance with the model parameter $\boldsymbol{J}$) can be applied to the rest channels. Consider the dynamics of pitch angular motion:

$$\begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\\ \dot{\tau}_y\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & \frac{1}{J_{yy}}\\ 0 & 0 & -\frac{1}{\alpha}\end{bmatrix}\begin{bmatrix}\theta\\ \dot{\theta}\\ \tau_y\end{bmatrix}+\begin{bmatrix}0\\ 0\\ \frac{1}{\alpha}\end{bmatrix}u_\theta+\begin{bmatrix}0\\ \frac{f_1}{J_{yy}}\\ f_2\end{bmatrix} \tag{12}$$

where f_1 and f_2 denote the lumped disturbance torque in the pitch motion and the lumped disturbance on the actuator, respectively. Note that the disturbance torque f_1 does not appear in the same channel as the torque command u_θ, which is known as mismatched disturbance. The challenge of rejecting mismatched disturbance does not lie in the disturbance estimation, but in the difficulty of designing an appropriate compensation signal in u_θ to cancel the effect of the disturbance from a different channel. To address this problem, a coordinate transformation is introduced to develop an auxiliary system that has matched disturbance and is equivalent to (12). Define an auxiliary variable

$$\eta = \tau_y + f_1, \tag{13}$$

which represents the total torques on the quadrotor, and then it is derived that

$$\begin{aligned}\dot{\eta} &= -\frac{1}{\alpha}\tau_y+\frac{1}{\alpha}u_\theta+f_2+\dot{f}_1\\ &= -\frac{1}{\alpha}\eta+\frac{1}{\alpha}u_\theta+f_2+\frac{1}{\alpha}f_1+\dot{f}_1.\end{aligned} \tag{14}$$

Dynamics (12) can be transformed into

$$\begin{bmatrix}\dot{\theta}\\ \ddot{\theta}\\ \dot{\eta}\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & \frac{1}{J_{yy}}\\ 0 & 0 & -\frac{1}{\alpha}\end{bmatrix}\begin{bmatrix}\theta\\ \dot{\theta}\\ \eta\end{bmatrix}+\begin{bmatrix}0\\ 0\\ \frac{1}{\alpha}\end{bmatrix}(u_\theta+\bar{f}), \tag{15}$$

where $\bar{f} = \alpha f_2 + f_1 + \alpha\dot{f}_1$ is the equivalent matched disturbance. This auxiliary system is called the transformed attitude control model, which is utilized to design the disturbance rejection mechanism in a more intuitive way.

To control a class of disturbed second-order systems in the form of (15), an UDE can be introduced into the classic PD controller to compensate for the effects of the lumped disturbances dynamically. Originally proposed in Zhong and Rees (2004), the underlying idea of UDE is that in the frequency domain, the lumped disturbance is approximated by passing it through a filter with an appropriate bandwidth. Using the same idea, when the desired state $\ddot{\theta}_d$, $\dot{\theta}_d$, and θ_d are given, the MUDE-based controllers in the following form are considered

$$u_\theta = \ddot{\theta}_d + k_p\tilde{\theta} + k_d\dot{\tilde{\theta}} + k_\eta\tilde{\eta} - \hat{\bar{f}}, \tag{16}$$

where

$$\tilde{\theta} = \theta_d - \theta, \quad \dot{\tilde{\theta}} = \dot{\theta}_d - \dot{\theta} \quad \tilde{\eta} = \eta_d - \eta, \tag{17}$$

and $\hat{\bar{f}}$ is the MUDE which estimates the lumped disturbance $\bar{f}$. Unfortunately, controller (16) is not directly applicable due to the following two facts. First, the state feedback $\tilde{\eta}$ is not available, since η contains an unknown disturbance f_1 and therefore η is not measurable. Second, $\hat{\bar{f}}$ cannot be derived from the transformed attitude control model (15), because of the unavailability of η.

To resolve these issues, the attitude control model (12) is used in what follows. Although $\bar{f}$ cannot be directly estimated, its estimation can still be constructed by separately estimating the three individual terms. Those estimates are respectively denoted by $\hat{f}_1$, $\hat{\dot{f}}_1$ and $\hat{f}_2$, and will be designed later. With $\hat{f}_1$ at hand,

$$\tilde{\eta} = \tau_d - \eta = \tilde{\tau} - \hat{f}_1, \tag{18}$$

and then the controller (16) can be modified as

$$u_\theta = \underbrace{\ddot{\theta}_d + k_p\tilde{\theta} + k_d\dot{\tilde{\theta}} + k_\eta\tilde{\tau}}_{\text{nominal control}} \underbrace{-\left[\left(k_\eta+1\right)\hat{f}_1 + \alpha\hat{\dot{f}}_1 + \alpha\hat{f}_2\right]}_{\text{MUDE}}, \tag{19}$$

where $\tilde{\tau} = \tau_d - \tau_y$ can be obtained by using the undisturbed version of (10) as a virtual sensor with the input u_θ, and τ_d is the desired torque and can be computed as

$$\tau_d = J_{yy}\ddot{\theta}_d. \tag{20}$$

The next step is to construct the MUDE in (19) by respectively designing estimators for f_1, $\dot{f}_1$ and f_2. First, similarly as the classic UDE, f_1 can be designed as

$$\hat{F}_1(s) = G_{f1}(s)F_1(s) = G_{f1}(s)\left[J_{yy}\ddot{\theta}(s) - \tau_y(s)\right], \tag{21}$$

and selecting

$$G_{f_1}(s) = \frac{1}{T^{f_1}s+1}, \ T^{f_1} > 0, \tag{22}$$

yields

$$\hat{F}_1(s) = \frac{s}{T^{f_1}s+1}\dot{\theta}(s) - \frac{1}{T^{f_1}s+1}\tau_y(s). \tag{23}$$

Second, the estimator for $\dot{f}_1$ is designed to satisfy

$$\hat{\dot{F}}_1(s) = G_{\dot{f}_1}(s)[\dddot{\theta}(s) - \dot{\tau}_y(s)]. \tag{24}$$

Since $\ddot{\theta}(s)$, $\dddot{\theta}(s)$ and $\dot{\tau}_y(s)$, which represent the angular acceleration, angular jerk and the derivative of the torque, are not measurable from the onboard sensors, the relative order of filter $G_{\dot{f}_1}(s)$ must be no

Table 2
Controller feedback gains.

CPID		UDE		MUDE			
k	6.0	k_p	2.0	k_p	2.0	$T_1^{\hat{f}_1}$	0.5
k_p	0.15	k_d	1.0	k_d	1.0	$T_2^{\hat{f}_1}$	1.0
k_i	0.2	T_f	0.1	k_η	1.0	T^{f_2}	0.5
k_d	0.003			T^{f_1}	0.05		

lower than 2 to ensure the physical realizability of (24). Therefore the following second-order filter is considered

$$G_{\hat{f}_1}(s) = \frac{1}{T_1^{\hat{f}_1} s^2 + T_2^{\hat{f}_1} s + 1},\ T_1^{\hat{f}_1} > 0,\ T_2^{\hat{f}_1} > 0, \tag{25}$$

which leads to

$$\hat{F}_1(s) = \frac{s^2}{T_1^{\hat{f}_1} s^2 + T_2^{\hat{f}_1} s + 1}\dot{\theta}(s) - \frac{s}{T_1^{\hat{f}_1} s^2 + T_2^{\hat{f}_1} s + 1}\tau_y(s). \tag{26}$$

Third, f_2 can be estimated by

$$\hat{F}_2(s) = G_{f_2}(s)\left[\dot{\tau}_y(s) + \frac{\tau_y(s)}{\alpha} - \frac{u_\theta(s)}{\alpha}\right], \tag{27}$$

where

$$G_{f_2}(s) = \frac{1}{T^{f_2} s + 1},\ T^{f_2} > 0, \tag{28}$$

is selected to avoid using the unavailable signal $\dot{\tau}_y(s)$. Combine (16), (27) and (28) yields

$$\begin{aligned}\hat{F}_2(s) =& \frac{1}{\alpha T^{f_2} s}[(\alpha s + 1)\tau_y(s) - \ddot{\theta}_d(s) - k_p\tilde{\theta}(s) - k_d\dot{\tilde{\theta}}(s) \\ & - k_\eta\tilde{\tau}_y(s) + \left(k_\eta + 1\right)\hat{F}_1(s) + \alpha\dot{\hat{F}}_1(s)].\end{aligned} \tag{29}$$

Then, the MUDE is constructed from the designs of (23), (26) and (29).

Remark 3. For the considered problem, the classic UDE-based controller (Lu et al., 2018) satisfies

$$u_\theta^{\mathrm{UDE}} = \underbrace{\ddot{\theta}_d + k_p\tilde{\theta} + k_d\dot{\tilde{\theta}}}_{\text{nominal control}} \underbrace{-\left[-\frac{1}{T_f}(k_d\tilde{\theta} + k_p\int_0^t\tilde{\theta}dt + \tilde{\theta})\right]}_{\text{UDE } \hat{f}_1}. \tag{30}$$

The MUDE is a variation of classic UDE in the sense that they share the same design philosophies: the frequency domain-based design and the use of appropriate filters to avoid using state derivative measurements. However, MUDE-based control is distinguished from the classic UDE-based control in several aspects. First, MUDE-based controller (19) additionally incorporates a torque feedback term $k_\eta\tilde{\tau}$ to enlarge the system stability margin and improve the transient performance. This modification is crucial for achieving high-accuracy attitude tracking when the quadrotor maneuvers aggressively. Second, the design of MUDE-based control is designed based on two different coordinates (12) and (15) to resolve the mismatched disturbance issue brought by the consideration of realistic actuators, whereas the classic UDE-based control is designed based on a simple disturbed double-integrator model (5). Third, the estimation accuracy of MUDE for disturbance f_1 is higher than that of classic UDE, because classic UDE-based control assumes ideal actuators and uses $u_\theta = \tau_y$ to derive $\hat{f}_1$, which leads to estimation errors from the disagreement between u_θ and τ_y in reality. Lastly, MUDE is capable of estimating not only the lumped disturbance f_1 on the attitude motion, but also the effects of time delay and disturbance (represented by f_2) on the actuators.

4. Stability and performance analysis

The tracking error dynamics are

$$\begin{bmatrix}\dot{\tilde{\theta}}\\ \ddot{\tilde{\theta}}\\ \dot{\tilde{\eta}}\end{bmatrix} = \boldsymbol{A}\underbrace{\begin{bmatrix}\tilde{\theta}\\ \dot{\tilde{\theta}}\\ \tilde{\eta}\end{bmatrix}}_{\tilde{\boldsymbol{x}}} - \boldsymbol{B}\left[\left(k_\eta + 1\right)\tilde{f}_1 + \alpha\left(\dot{\tilde{f}}_1 + \tilde{f}_2\right)\right], \tag{31}$$

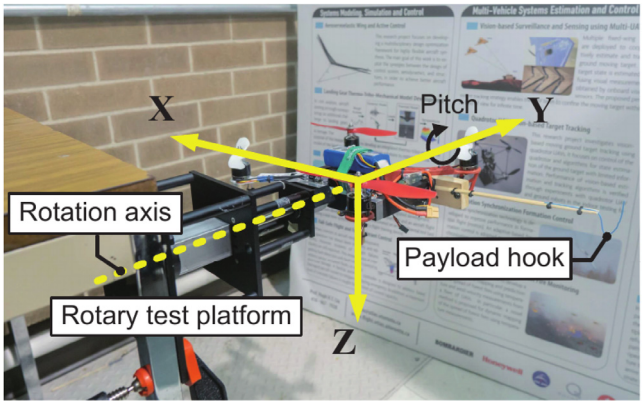


Fig. 7. The rotary test platform for quadrotor attitude control.

Table 3
Case studies in the experiments.

Cases	Scenarios	Disturbance	Parameter tuning	Tested controllers
1	Hovering & sinusoidal trajectory tracking	×	–	(48), (30), (19)
2	Step trajectory tracking	×	–	(48), (30), (19)
3	Hovering	✓	–	(48), (30), (19)
4	Sinusoidal trajectory tracking	✓	–	(48), (30), (19)
5	Hovering	✓	T^{f1}	(19)
6	Hovering & sinusoidal trajectory tracking	×	T^{f2}	(19)
7	Remotely piloted flight	✓	–	(19)

where

$$\boldsymbol{A} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & \frac{1}{J_{yy}}\\ -\frac{k_p}{\alpha} & -\frac{k_d}{\alpha} & -\frac{(k_\eta+1)}{\alpha}\end{bmatrix},\quad \boldsymbol{B} = \begin{bmatrix}0\\ 0\\ \frac{1}{\alpha}\end{bmatrix}, \tag{32}$$

$$\tilde{f}_1 = f_1 - \hat{f}_1,\quad \dot{\tilde{f}}_1 = \dot{f}_1 - \dot{\hat{f}}_1,\quad \tilde{f}_2 = f_2 - \hat{f}_2. \tag{33}$$

The characteristic polynomial of $\boldsymbol{A}$ is

$$|s\boldsymbol{I}_3 - \boldsymbol{A}| = s^3 + \frac{k_\eta + 1}{\alpha}s^2 + \frac{k_d}{J_{yy}\alpha}s + \frac{k_p}{J_{yy}\alpha}. \tag{34}$$

By Routh criterion, the stability conditions are

$$k_p > 0,\ k_d > 0,\ k_\eta > -1,\ \left(k_\eta + 1\right)k_d - \alpha k_p > 0. \tag{35}$$

Under the stability conditions, the tracking accuracy relies on the estimation residual error

$$\tilde{f}_{res} = \left(k_\eta + 1\right)\tilde{f}_1 + \alpha\left(\dot{\tilde{f}}_1 + \tilde{f}_2\right). \tag{36}$$

It is seen that $\alpha\left(\dot{\tilde{f}}_1 + \tilde{f}_2\right)$ has little impact on the tracking accuracy since the time constant of the motor $\alpha \approx 0.04$ is small. Therefore the bandwidths of estimators (which determines the noise sensitivity as well) for $\dot{f}_1$ and f_2 are not necessarily to be too large. The estimation errors satisfy

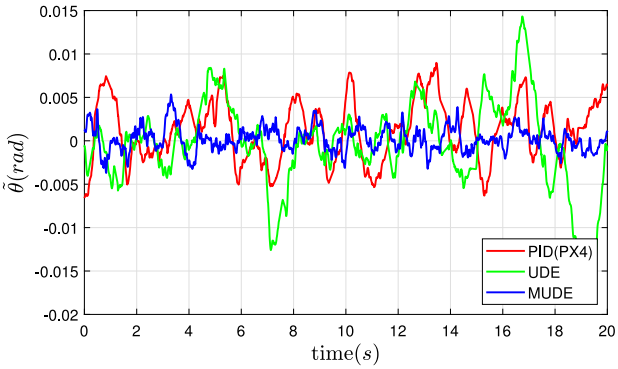
$$\tilde{F}_1(s) = \left[1 - G_{f_1}(s)\right]F_1(s) = \frac{T^{f_1}s}{T^{f_1}s + 1}F_1(s), \tag{37}$$

$$\tilde{\dot{F}}_1(s) = \left[1 - G_{\hat{f}_1}(s)\right]\dot{F}_1(s) = \frac{T_1^{\hat{f}_1}s^2 + T_2^{\hat{f}_1}s}{T_1^{\hat{f}_1}s^2 + T_2^{\hat{f}_1}s + 1}\dot{F}_1(s), \tag{38}$$

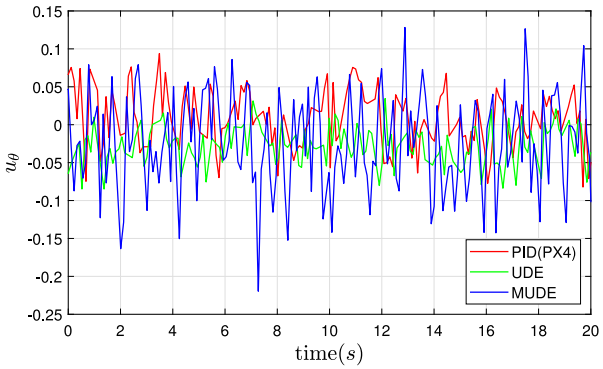
$$\tilde{F}_2(s) = \left[1 - G_{f_2}(s)\right]F_2(s) = \frac{T^{f_2}s}{T^{f_2}s + 1}F_2(s). \tag{39}$$

To show how the estimation errors are related to the filter parameters, the following parameter mapping is applied

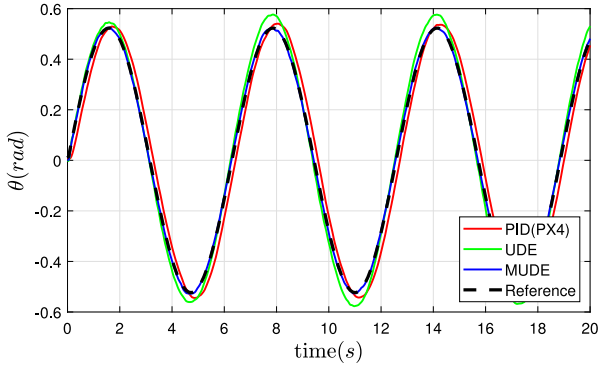
$$c_0 = \varepsilon\alpha_1, c_1 = \varepsilon\alpha_2, \ldots, c_k = \varepsilon\alpha_k, \tag{40}$$



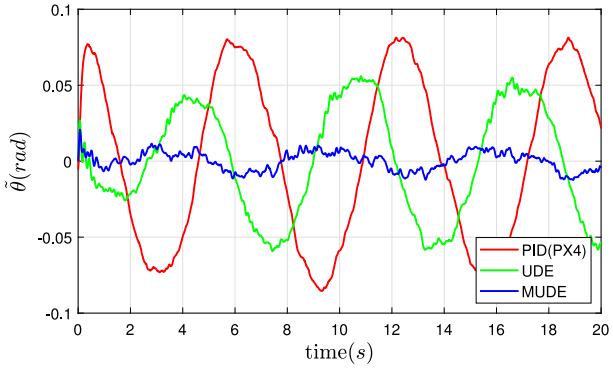
(a) The pitch angle tracking errors (hovering).



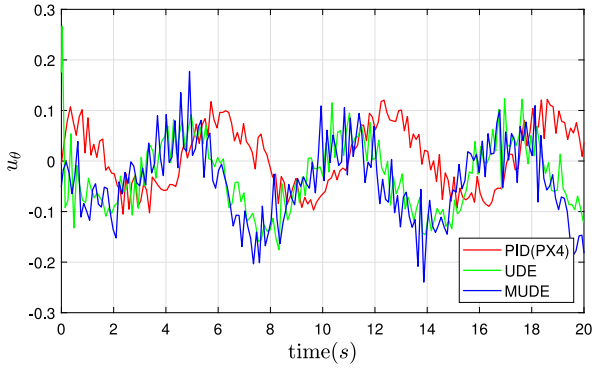
(b) The torque commands (hovering)



(c) The pitch angle trajectories (sinusoidal trajectory tracking)



(d) The pitch angle tracking errors (sinusoidal trajectory tracking)



(e) The torque commands (sinusoidal trajectory tracking).

Fig. 8. Experimental results for Case 1.

to the general form of stable low-pass filters:

$$G(s) = \frac{1}{c_0 s^{k+1} + \cdots + c_k s + 1}, \tag{41}$$

where $\alpha_1, \ldots \alpha_{k+2}$ and ε are positive parameters. Then the following results can be obtained.

Lemma 1. *For all stable filters in the form of (41) with parameters replaced by (40), letting $\varepsilon \to 0$ leads to sufficiently small $\tilde{F}(s) = [1 - G(s)]F(s)$ provided that $F(s)$ is bounded.*

Proof. By (40) and (41), it readily has

$$\tilde{F}(s) = \varepsilon \cdot \frac{\alpha_1 s^{k+1} + \cdots + \alpha_{k+1} s}{\varepsilon\alpha_1 s^{k+1} + \cdots + \varepsilon\alpha_{k+1} s + 1} F(s). \tag{42}$$

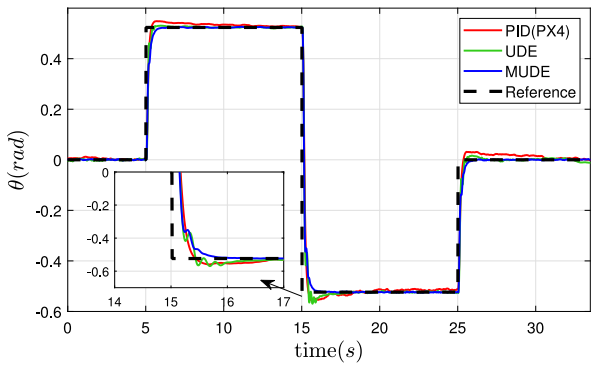
Note that mapping (40) will not change the poles of filter $G(s)$. Therefore $\frac{\alpha_1 s^{k+1} + \cdots + \alpha_{k+1} s}{\varepsilon\alpha_1 s^{k+1} + \cdots + \varepsilon\alpha_{k+1} s + 1}$ is bounded-input-bounded-output stable since its denominator is identical to that of $G(s)$. Then by the boundedness of $F(s)$, it is clear that $\varepsilon \to 0$ renders $\tilde{F}(s)$ arbitrarily small. The proof is complete.

Lemma 1 concludes that the estimation errors of the proposed UDEs $\hat{F}_1$, $\hat{\dot{F}}_1$ and $\hat{F}_2$, can be reduced by tuning the filter parameters. This result will be used to show the ultimate bounds of the estimation residual error (36) and tracking error.

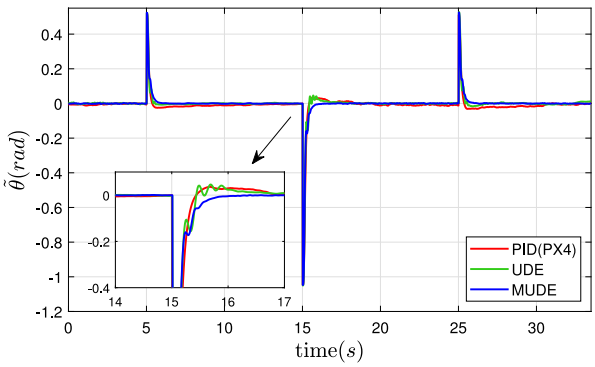
Theorem 1. *Under Assumption 1 and stability conditions (35), when the following parameter mapping*

$$T^{f_1} = \varepsilon\alpha_1, T_1^{\hat{f}_1} = \varepsilon\alpha_2, T_2^{\hat{f}_1} = \varepsilon\alpha_3, T^{f_2} = \varepsilon\alpha_4, \quad \alpha_1, \alpha_2, \alpha_3, \alpha_4, \varepsilon > 0, \tag{43}$$

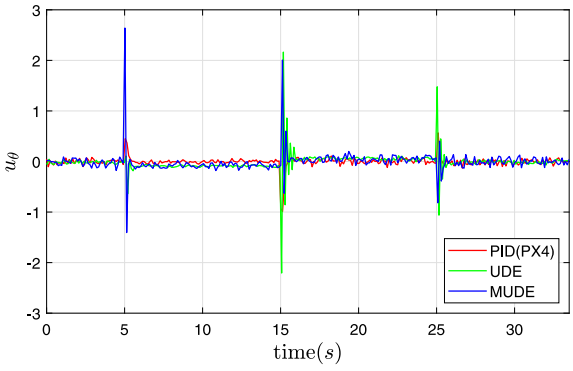
is applied to filters $G_{f_1}(s)$, $G_{\hat{f}_1}(s)$ and $G_{f_2}(s)$, the estimation residual error (36) is bounded by $f_b(\varepsilon)$, and the tracking error $\|\tilde{\mathbf{x}}(t)\|_2$ is ultimately bounded by $\bar{f}_b(\varepsilon)$ as $t \to \infty$, where the bounds satisfies $f_b(\varepsilon) \to 0$ and $\bar{f}_b(\varepsilon) \to 0$ as $\varepsilon \to 0$.



(a) The pitch angle trajectories.



(b) The pitch angle tracking errors.



(c) The torque commands.

Fig. 9. Experimental results for Case 2.

Proof. Assumption 1 guarantees the boundedness of $F_1(s)$, $\dot{F}_1(s)$ and $F_2(s)$. Applying the result of Lemma 1 to estimation error dynamics (37)–(39) straightforwardly yields $\tilde{F}_{res}(s) = \varepsilon H(s)$, where

$$H(s) = \frac{\alpha_1 s}{\varepsilon\alpha_1 s + 1} F_1(s) + \frac{\alpha_2 s^2 + \alpha_3 s}{\varepsilon\alpha_2 s^2 + \varepsilon\alpha_3 s + 1} \dot{F}_1(s) + \frac{\alpha_4 s}{\varepsilon\alpha_4 s + 1} F_2(s) \tag{44}$$

is a bounded signal. Therefore in the time domain it can conclude that

$$\tilde{f}_{res}(t) \le \varepsilon \|h(t)\|_\infty \triangleq f_b(\varepsilon), \tag{45}$$

where $f_b(\varepsilon) \to 0$ as $\varepsilon \to 0$. Under stability conditions (35), it is clear that the zero-input response of (31) converges to the origin, i.e. $\|\tilde{\boldsymbol{x}}_{zir}(t)\|_2 = \|e^{\boldsymbol{A}t}\tilde{\boldsymbol{x}}(0)\|_2 \to 0$ as $t \to \infty$. The 2-norm of the zero-state response satisfies

$$\begin{aligned} \|\tilde{\boldsymbol{x}}_{zsr}(t)\|_2 &= \|\textstyle\int_0^t e^{\boldsymbol{A}(t-t_s)}\boldsymbol{B}\tilde{f}_{res}(t_s)dt_s\|_2 \\ &\le \|\textstyle\int_0^t e^{\boldsymbol{A}(t-t_s)}dt_s\|_2 \|\boldsymbol{B}\|_2 \|\tilde{f}_{res}(t)\|_\infty \\ &\le \|(e^{\boldsymbol{A}t} - \boldsymbol{I}_3)\|_2 \|\boldsymbol{A}^{-1}\|_2 \|\boldsymbol{B}\|_2 \|\tilde{f}_{res}(t)\|_\infty \end{aligned} \tag{46}$$

By (45) and (46), as $t \to \infty$ results

$$\|\tilde{\boldsymbol{x}}(t)\|_2 \le \|\tilde{\boldsymbol{x}}_{zir}(t)\|_2 + \|\tilde{\boldsymbol{x}}_{zsr}(t)\|_2 \le \|\boldsymbol{A}^{-1}\|_2 \|\boldsymbol{B}\|_2 f_b(\varepsilon) \triangleq \bar{f}_b(\varepsilon). \tag{47}$$

This ends the proof.

Theorem 1 shows that decreasing ε (which, correspondingly, reduces filter parameters T^{f_1}, $T_1^{f_1}$, $T_2^{f_1}$ and T^{f_2}) renders the steady-state estimation errors and closed-loop tracking error arbitrarily small. This result provides a clear relationship between the filter parameters and tracking accuracy, making the parameter tuning easier.

Remark 4. In practice, however, there is a contradiction between rejecting disturbances and reducing the sensitivity of MUDE to measurement noises (Zhu, Zhu, Liu, & Qin, 2020). The measurement noises in θ and $\dot{\theta}$ due to the low-cost onboard sensors, and the τ_y estimation error caused by the actuator modeling and identification error, may together limit the smallest feasible ε. Therefore, a trade-off needs to be made between improving the tracking performance and reducing the sensitivity to measurement noise and actuator model uncertainties.

5. Experimental results

5.1. Experimental setup

A quadrotor prototype composed of an F330 DJI frame, PropDrive 2826 1200Kv motors, LitteBee 20A ESCs, 8045 propellers, and a 3S 2200 mAh 25C LIPO battery is used for the experiments.

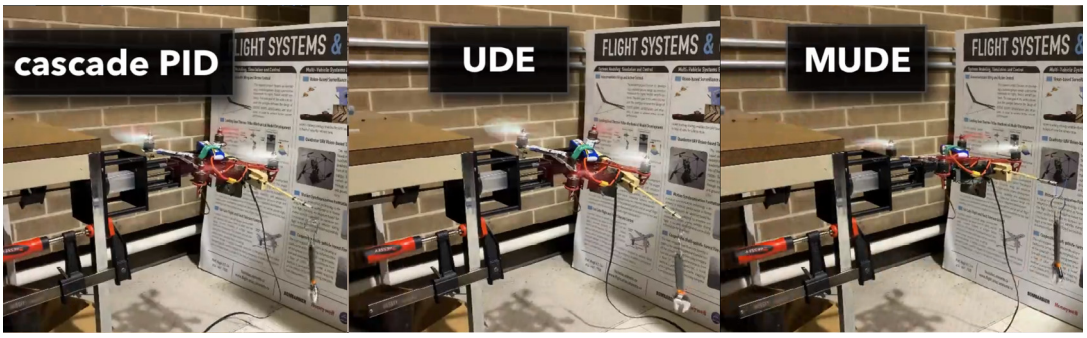
The experiments are conducted on the rotary test platform shown in Fig. 7. The platform's base is mounted on the table, and the quadrotor is installed on the rotatable end of the platform. The y-axis of the quadrotors body-frame is aligned with the rotation axis of the platform, such that only the pitch motion of the quadrotors is activated. A bamboo stick with a payload hood is attached to the quadrotor, and during the experiments, a payload can be hung on the hook to simulate a possibly time-varying external disturbance torque on the pitch motion.

5.2. Tested controllers

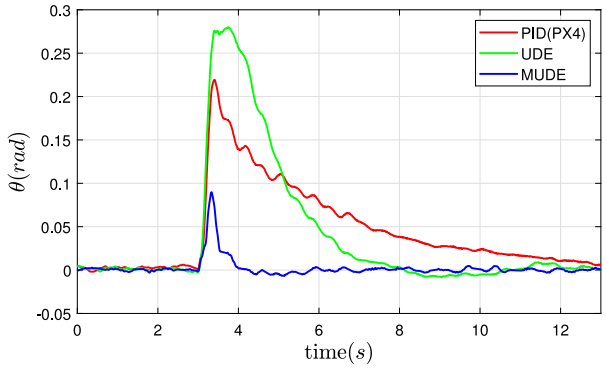
In the experiments, three controllers are tested, including the cascade PID (CPID) controller (Dronecode, 2020), the classic UDE-based controller (Lu et al., 2018) given by (30), and the proposed MUDE-based controller (19). The cascade PID controller is embedded inside PX4 firmware and is widely used in most commercial autopilots. It is designed in a cascade form as follows

$$u_\theta^{\text{CPID}} = k_p\tilde{\theta} + k_i\textstyle\int_0^t\tilde{\theta}dt + k_d\dot{\tilde{\theta}}, \tag{48}$$

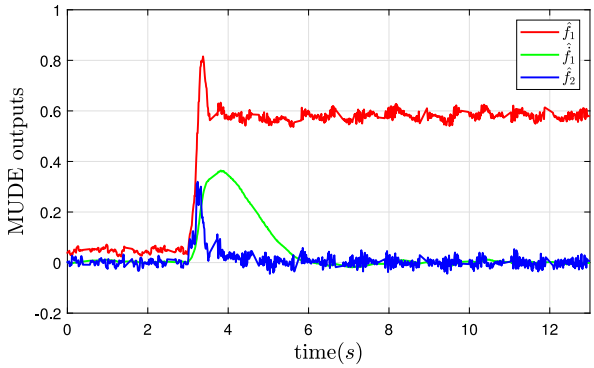
where the reference for the attitude angular rate $\dot{\theta}_d$ is generated by a proportional controller $\dot{\theta}_d = k\tilde{\theta}$.



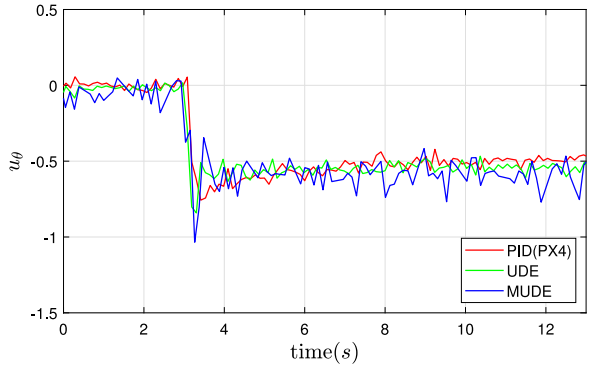
(a) The photos shot at $t = 3$ s during the experiment.



(b) The pitch angle trajectories.



(c) The estimation result.



(d) The torque commands.

Fig. 10. Experimental results for Case 3.

Remark 5. In the proposed MUDE-based controller, the uncertainty and disturbance estimators (23)(26)(29) are designed in the frequency domain. To program it in C language and implement the whole solution on a Pixhawk flight controller running PX4 firmware 1.8.2, they need to be translated into the discrete-time form. The computational complexity of the proposed solution is considered to be low in the sense that the Pixhawk only equips with an STM32F427 processor running at 168 MHz. The codes are published in Github and can be found at https://github.com/potato77/Firmware_MUDE.

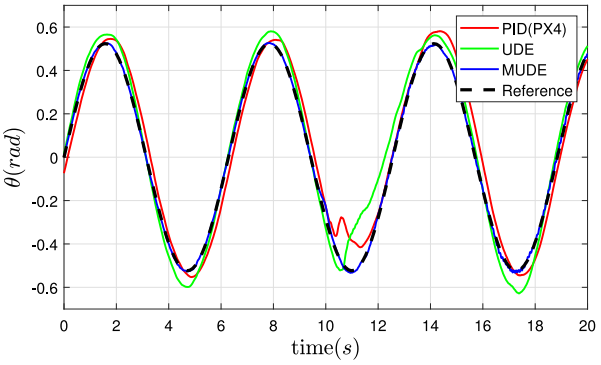
For the aforementioned feedback controllers investigated in the experiments, the pitch angle is obtained from the embedded EKF module in PX4 firmware, while the pitch rate is measured by the gyroscope and is processed by a low-pass filter. The default controller feedback gains are shown in Table 2. It is noteworthy that the same control feedback gains k_p and k_d are chosen for classic UDE and MUDE to give an equitable comparison result. The flight-proven feedback gains are selected for CPID.

5.3. Experimental results

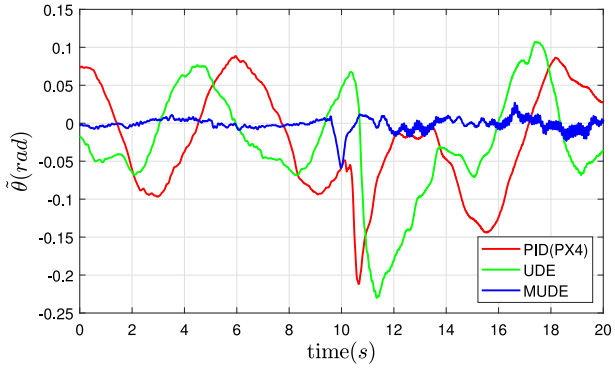
The case studies considered in the experiments are summarized in Table 3. Three typical references are considered, and the sinusoidal reference satisfies

$$\theta_d = 0.5236 \cdot \sin(4t) \text{ rad}, \tag{49}$$

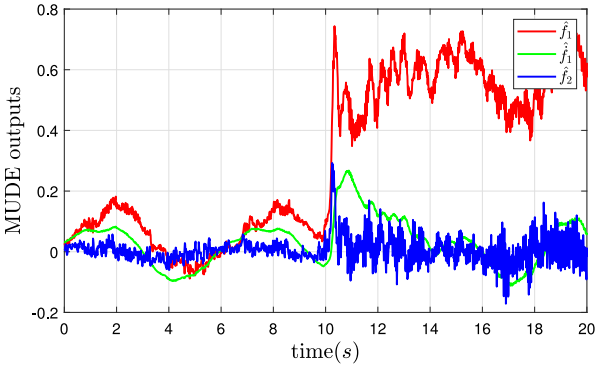
where 0.5236 rad = 30° is almost the maximum tilt angle of the quadrotor during a real flight for safety considerations. The disturbance is added to the quadrotor pitch motion by hanging a payload (150 g) on the hook, which, in the steady-state when $\theta = 0$, is an equivalent disturbance torque of approximately 0.6 N m. When the pitch angle varies, the payload acts as a time-varying disturbance ($0.6\cos\theta$ N m, when neglecting the swinging of the payload) to the system. In Cases 1–4, the CPID controller (48), the UDE-based control (30) and the proposed MUDE-based control (19) are tested and compared. In Cases



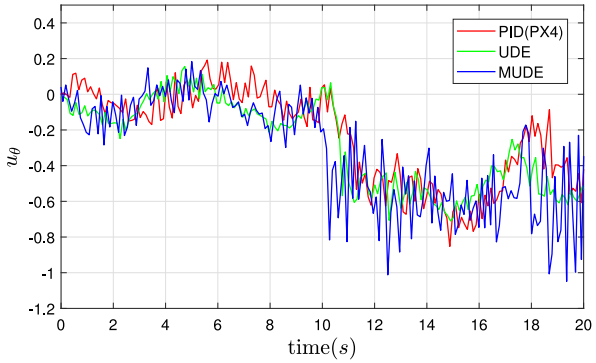
(a) The pitch angle trajectories.



(b) The pitch angle tracking errors.

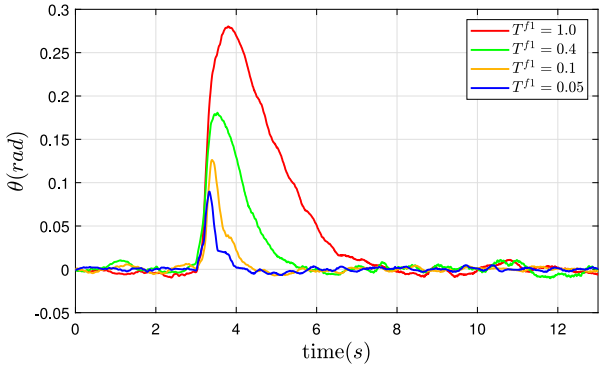


(c)The disturbance estimation result.

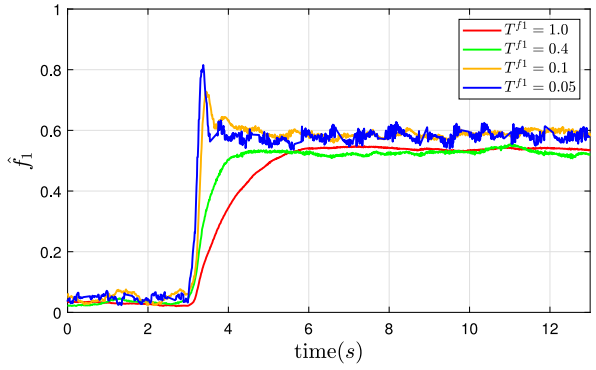


(d) The torque commands.

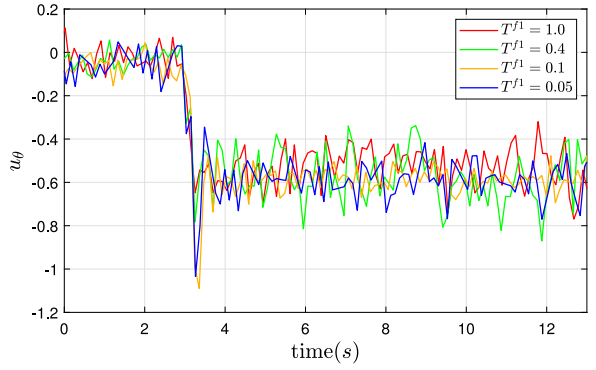
Fig. 11. Experimental results for Case 4.



(a) The pitch angle trajectories under different T^{f1}.

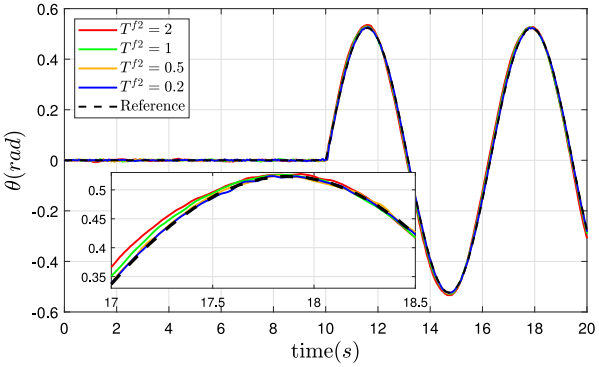


(b) The estimation result of f_1 under different T^{f1}.

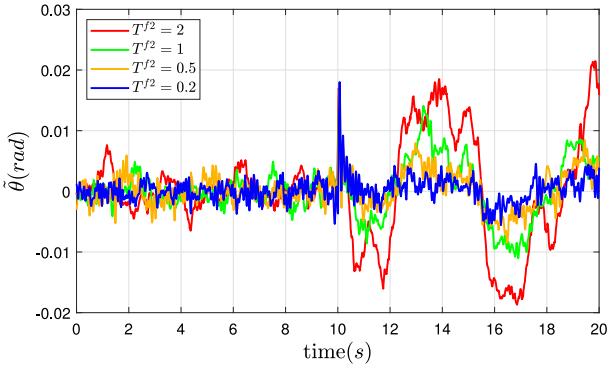


(c) The torque commands.

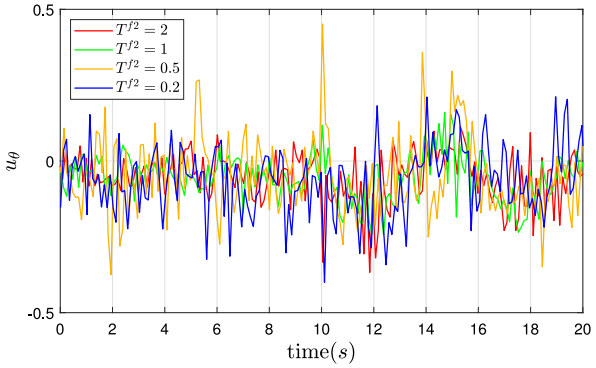
Fig. 12. Experimental results for Case 5.



(a) The pitch angle trajectories under different T^{f2}.



(b) The pitch angle tracking errors under different T^{f2}.



(c) The torque commands.

Fig. 13. Experimental results for Case 6.

5–6, the parameters T^{f1} and T^{f2} are respectively tuned for MUDE, demonstrating the effectiveness of the proposed tuning guidelines. In Case 7, the quadrotor is manually controlled to fly in an indoor environment, and the real flight test results are presented.

The videos of the experiments can be found at https://www.bilibili.com/video/av60962814/ (for Cases 1–6) and https://www.bilibili.com/video/BV1v54y1q7X7 (for Case 7).

5.3.1. Case 1

The results are shown in Fig. 8. It is noticed that the tracking errors of three controllers are close, but MUDE-based control delivers smallest tracking error when the quadrotor is tracking a constant reference signal, as shown in Fig. 8(a). Since the quadrotor is in the steady-state, the torque command u_θ converges to a small neighborhood of zero, and thus the motors are spinning at a nearly constant speed shown in Fig. 8(b). Under this particular circumstance, the dynamics of the actuators can be ignored.

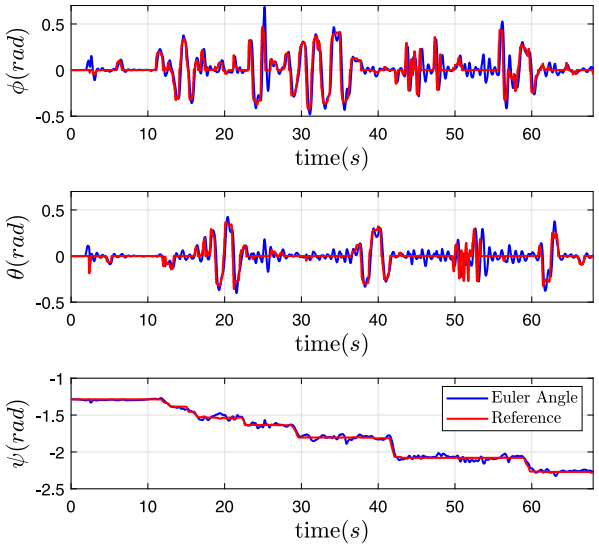
As for the sinusoidal trajectory tracking results shown in Fig. 8(c)–(e), however, there is a remarkable improvement in the tracking accuracy by the MUDE-based controller, while the results of its rival controllers exhibit some residual errors of an amplitude of around $0.05 \sim 0.08$ rad. This is because the actuator-generated torque is not in agreement with the torque command due to the dynamics of the actuator and the time delay, when the quadrotor is doing maneuvers. Compared with CPID and classic UDE-based control, the term $\hat{f}_2$ in the MUDE actively compensates for this torque difference, and the additional torque error feedback term $k_\eta \tilde{\tau}$ in the proposed controller (19) helps to enlarge the system stability margin and improve the convergence speed of attitude tracking, thus overall steady-state and transient tracking performance is simultaneously enhanced. This point is verified by the torque commands generated by these controllers, as shown in Fig. 8(e). CPID lacks feedforward control, and thus the torque command has an obvious phase delay, whereas the feedforward term in the classic UDE-based control helps to correct this phase delay to some extent. The MUDE additionally compensates for the dynamic delay and time delay of the actuator, which further corrects the phase delay in the torque command. Based on the above analysis, with regard to accurate attitude tracking problems, especially when the quadrotor maneuvers aggressively, the control design considering actuator dynamics are of practical importance. Similar conclusions can be found in Castillo et al. (2019), in which the considered actuator model is a first-order system. It will be showed in Case 6 that the time delay of the actuator considered in this paper has a significant impact on the tracking performance, and thus the proposed approach has certain advantages over the DOBC proposed in Castillo et al. (2019) in the sense that the actuator delay is well-address by the MUDE-based control.

5.3.2. Case 2

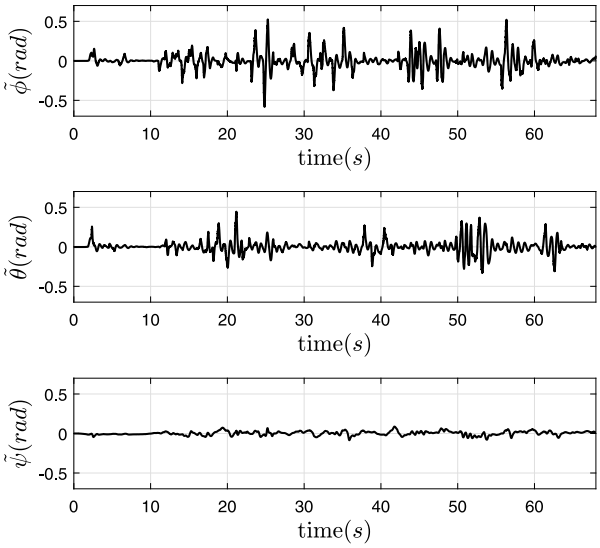
The step trajectory tracking, which is very common in real flight missions, is investigated in this case. The results are shown in Fig. 9. Note that since the step reference is not differentiable with respect to time, the angular rate references for the proposed controller and classic UDE-based controllers are calculated by the same proportional controller $\dot{\theta}_d = k\tilde{\theta}$ as CPID, while the feedforward control and the torque reference for MUDE-based control are set to zero. It is seen that the tracking error of MUDE-based control exhibits no overshoot, fast convergence, small oscillation, and very small steady-state error. In contrast, CPID and UDE-based control cause different levels of overshoot and recover slower than MUDE-based control. CPID is the best among all the tested controllers in terms of oscillations, and classic UDE suffers from obvious oscillation during the transition.

5.3.3. Case 3

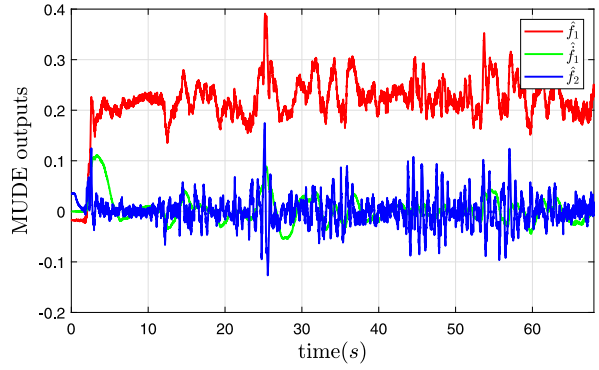
In this case, the constant disturbance rejection performance of different controllers is considered. The quadrotor hovers at $t = 0 \sim 3$ s



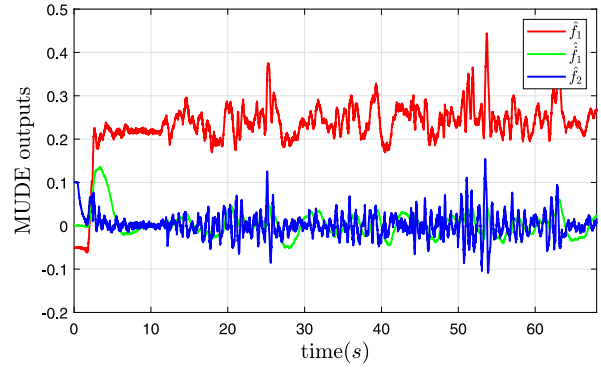
(a) The euler angle trajectories.



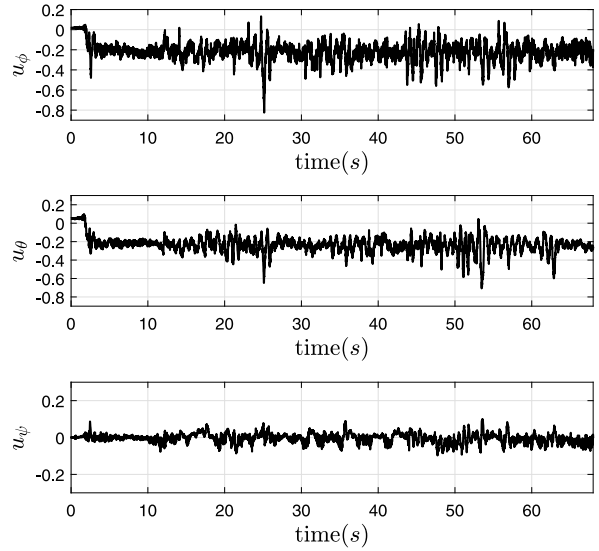
(b) The euler angle tracking errors.



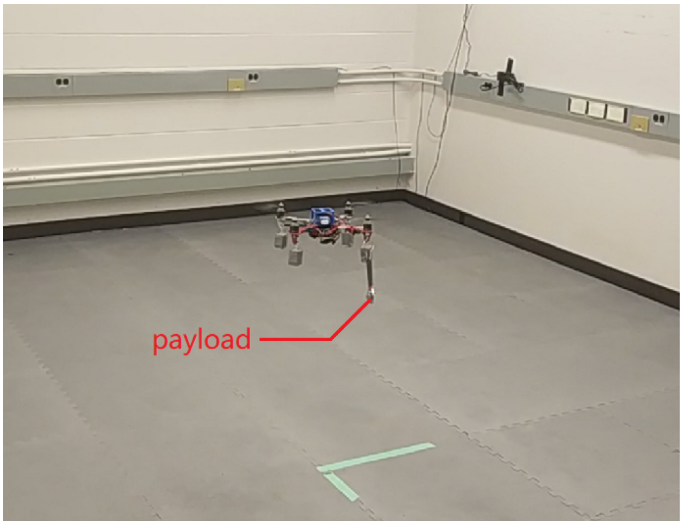
(c) The X-axis disturbance estimation result.



(d) The Y-axis disturbance estimation result.



(e) The torque commands.



(f) The suspended payload in the experiment.

Fig. 14. Experimental results for Case 7.

and the payload is hung to the hook at $t = 3$ s. The results are shown in Fig. 10. It can be observed from screenshot in Fig. 10(a) and (b) that the MUDE-based control is least disturbed by the disturbance, and it recovers its tracking performance faster than the other two controllers. Moreover, Fig. 10(c) shows the three estimation terms generated by MUDE. It is seen that $\hat{f}_1$ quickly converges to 0.6 when the quadrotor

is disturbed by a nearly constant external torque 0.6 N m, while $\hat{f}_1$ and $\hat{f}_2$ converge to 0. These results are reasonable since f_1 and f_2 are close to zero under the steady-state conditions.

5.3.4. Case 4

In Case 4, the tracking performance of three controllers under a time-varying disturbance is investigated. The experiment starts with the quadrotor tracking a sinusoidal trajectory, and then the payload is hung to the hook at around $t = 10$ s. As shown in Fig. 11(a)–(b), in the steady-state ($t = 0 \sim 10$ s) the proposed MUDE-based control delivers almost perfect tracking of the sinusoidal reference. As for the disturbance rejection, the MUDE-based control is able to deliver fast tracking performance recovery and performs the best among the three controllers in terms of the steady-state tracking accuracy.

From the experimental results of Case 1–4, it is seen that MUDE-based control can not only reject various disturbances on the system but also deliver high tracking accuracy during maneuvers.

5.3.5. Case 5

This case study illustrates the effectiveness of tuning T^{f1} for MUDE to deliver better external disturbance torque rejection performance. The experimental setup is the same as Case 3. Four different parameters are selected respectively as $T^{f1} = \{1, 0.4, 0.1, 0.05\}$, and their correspond results are shown in Fig. 12(a). Reducing T^{f1} improves the disturbance rejection performance, as analyzed in . More specifically, a smaller T^{f1} delivers a smaller deviation from the steady-state when the disturbance is added, and can estimate and compensate for the disturbance in a shorter time. However, T^{f1} cannot be arbitrarily small in practice due to measurement noises. From (23), a small T^{f1} can significantly amplify the high-frequency noise in the angular rate measurement $\dot{\theta}$, which exactly explains the chatterings in the $\hat{f}_1$ shown in Fig. 12(b) when T^{f1} is chosen to be smaller than 0.1. Although the chattering can be largely filtered out by the damping in the quadrotor and the actuator dynamics, it may still lead to actuator wear or even system instability. Therefore, T^{f1} should be carefully chosen in applications.

5.3.6. Case 6

The results of tuning parameter T^{f2} are demonstrated in Fig. 13(a)–(c). The quadrotor hovers at $t = 0 \sim 10$ s and tracks a sinusoidal trajectory after $t = 10$ s. T^{f2} is the parameter for the estimator (29) which is designed to estimate the time delay and disturbance term $\boldsymbol{f}_2 = \frac{1}{\alpha}[\boldsymbol{u}_\tau(t-\tau_0) - \boldsymbol{u}_\tau(t)] + \boldsymbol{d}_\tau$ on the actuators. It can be observed that during hovering, tuning T^{f2} has insignificant influence on the tracking accuracy. This is reasonable because under the steady-state condition, the motors spin at almost a constant speed (which is a straightforward result of steady torque commands). Therefore the lumped disturbance $\boldsymbol{f}_2$ is largely determined by the input disturbance $\boldsymbol{d}_\tau$ on the actuators, which is, in most cases, negligible. However, when tracking a time-varying reference ($t = 10 \sim 20$ s), since the torque command $\boldsymbol{u}_\tau$ is no longer constant, the effect of $\boldsymbol{f}_2$ becomes significant due to the model uncertainty term $\frac{1}{\alpha}[\boldsymbol{u}_\tau(t-\tau_0) - \boldsymbol{u}_\tau(t)]$. Thus when T^{f2} is reduced from 2 to 0.2, a remarkable improvement in the tracking performance is observed. This result implies that the time delay of the actuator cannot be ignored and tuning T^{f2} is of importance especially for aggressive maneuvers.

5.3.7. Case 7

This case is carried out to evaluate the disturbance rejection and tracking performance of the proposed MUDE-based control in a real flight. The quadrotor is set to manual flight mode. Random and aggressive roll/pitch commands (i.e., euler angle references) are given through a remote controller by a pilot. A suspended payload of 150 g is attached to one of the landing gears (see Fig. 14(f)), which together with the coupling dynamics of the quadrotor due to the asymmetric mass distribution, can be regarded as a time-varying external disturbance torque added to the quadrotor. The flight test results are shown in Fig. 14(a)–(e). Generally speaking, MUDE-based control is capable of rejecting the disturbance and delivering good attitude tracking performance. Specifically, since the reference is random and might change suddenly in this case, the tracking errors exhibit similar peaks to those in Case 2, as shown in Fig. 14(b). Despite these peaks, MUDE-based control could quickly eliminate tracking errors. Fig. 14(c)–(d) demonstrate the disturbance estimation results for the pitch and roll axes, from which it is seen that the external disturbance torques produced by the suspended payload are dynamically estimated by the MUDE. With these estimation signals, the controller could actively compensate for the disturbances, which not only guarantees the flight safety under vast disturbances but also achieves fast and accurate tracking of the reference.

6. Conclusion

An MUDE-based robust controller is proposed for quadrotor accurate attitude tracking and external disturbance rejection. It is shown that the tracking performance of classic UDE-based schemes, especially during aggressive maneuvers, can be improved by a systematic consideration of the actuator dynamics in both the design of the nominal controller and UDE. The immeasurable states of the actuators, i.e., the actuator-generated torques, are estimated using the identified FOPTD model. By tuning a single parameter of MUDE, the attitude tracking and disturbance estimation errors can be significantly reduced, which is experimentally verified on a real quadrotor platform.

Declaration of competing interest

This work was supported in part by the National Natural Science Foundation of China under Grant 61773095 and Grant 61973327, and in part by the China Scholarship Council (CSC) under Grant 201706070080.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.conengprac.2020.104721. The video for the quadrotor attitude control experiments on the rotary test platform can be found at https://www.bilibili.com/video/av60962814/, and the video for the indoor quadrotor attitude control flight test with a suspended payload can be found at https://www.bilibili.com/video/BV1v54y1q7X7.

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